

The Integer Web Principle

Casey Koons (vision-derived structural insight) + Keeper (formalization)

2026-05-20 Wednesday EOD

The Integer Web Principle

Casey-named statement (working hypothesis register, per Cal Calibration #13)

“The BST framework operates AS IF substrate’s boundaries are the BST primary integers themselves. Each integer APPEARS TO hold a web of operations, identities, and observable manifestations. The webs APPEAR TO interconnect at Type C convergence patterns. Cross-web intersections produce overdetermined identities (Graph Forces diagnostic). Substrate engineering MAY BE understood as operating at web intersections, not at substrate interior; each apparatus design appears to select which web-node to expose. The substrate’s complete structure MAY BE the braided union of these 6 (or 8 with observer + temporal) integer-webs.”

This is a WORKING ONTOLOGICAL HYPOTHESIS, not a derived structural claim. The “appears to” register is load-bearing: BST does not claim integer-as-boundary by mathematical proof; it observes that the framework operates coherently under this ontological reading.

What the framework observes (six BST primary integer-webs)

Each BST primary integer appears to function as the central node of a web of operations and observable manifestations:

Integer	Web (observable manifestations)
rank = 2	Duality structures — binary, parity, observation pairing, two-channel measurement, Hopf-1 winding, real rank of D_{IV}^5

Integer	Web (observable manifestations)
$N_c = 3$	Color structures — three-color SU(3), three-generation, triadic relationships, three-quark trefoil, codimension in Shilov boundary
$n_C = 5$	Five-fold structures — complex 5-dim manifold, pentamorous nuclear shell, Klein $A_5 \subset SO(5)$, generation count
$C_2 = 6$	Casimir/Lie structures — Lichnerowicz scalar 75/4, top Chern $c_5(Q^5)=6$, Casimir eigenvalue, group invariants, K3 hyperkähler
$g = 7$	Genus structures — Bergman genus, $GF(2^g)=GF(128)$ cyclotomic, Mersenne $M_g=127$, Casimir asymmetric ratio, genus invariant of Q^5
$N_{max} = 137$	Fine-structure/spectral structures — α , Haldane exclusion, channel capacity, BaTiO3 137-plane, spectral ceiling

Plus 2 substrate-external integer-webs (per Casey vision Wednesday afternoon): - Observer-BC web: substrate-coupling rate α_{CI} , measurement protocol, attention gradient - Temporal-BC web: Koons tick $t_{substrate}$, commitment-cycle granularity, initial conditions

Total: 6 substrate-internal integer-webs + 2 substrate-external = 8 integer-webs in the framework's structural ontology.

Bulk-boundary 2-face refinement (Casey vision)

Each integer-web has TWO FACES at the commitment-circle level: - Bulk face: inside the commitment circle disk (D^2 bounded by S^1) - Boundary face: on the S^1 perimeter

The Casimir asymmetric ratio = g lives at g -web boundary-face. The BaTiO3 137-plane lives at N_{max} -web boundary-face. Eigentone resonance lives at bulk-face (resonance with bulk-interior 2D continual reorganization).

4-Zone commitment cycle refinement (Casey vision)

Each commitment cycle has FOUR structural zones, integer-web manifestations distribute across them:

1. Inner edge (absorption boundary, ACTIVE): substrate boundary where info enters

2. Bulk interior (RECORDING + CONTINUAL 2D SEMI-CHAOTIC REORGANIZATION): substrate's "thinking" happens here
3. Between edges (LIGHT EMISSION zone): where output crosses from substrate to observable
4. Outer edge (emission boundary, ACTIVE): substrate boundary where info exits

Each integer-web has manifestations in multiple zones. Per Elie Toy 3154 apparatus-zone mapping (Wednesday Phase 3): 8 BST experimental apparatus map to specific zones — Casimir/BaTiO₃/Hg-1223 → Zone 4 active; Lamb/BCS → Zone 2 bulk; Bell → Zone 3 emission; commitment manipulation → Zone 1 absorption.

Three-scale substrate operation (Casey vision)

Integer-webs operate at THREE scales: - Intra-cycle (Scale 1): 4-zone structure within one commitment circle - Inter-cycle local (Scale 2): integer-edges connect adjacent commitment circles in 2D contact - Inter-cycle long-distance (Scale 3): integer-edges maintain non-local correlations across distant cycles (substrate of cognition per Substrate Cognition Hypothesis L2-cognition sub-class)

Falsifier specification

The principle is falsified by: - Discovery of a BST primary integer whose appearances DON'T form a coherent web (random, disconnected, no Type C convergence pattern visible) - Discovery of substrate observables that DON'T align with any of the 6 (or 8) integer-webs - Failure of the Graph Forces diagnostic to find overdetermined-identity clustering at integer-web intersections at rates significantly above null model

Specifically: if a comprehensive scan of BST observables finds $\geq 10\%$ that cannot be associated with any integer-web, the framework's ontological claim weakens significantly.

What the framework does NOT claim

- NOT a mathematical proof that integers ARE boundaries in some classical-math sense: this is structural ontology, not topology theorem
- NOT exhaustive enumeration of integer-webs: 6 (or 8) is current observational state; refinements possible as substrate work matures
- NOT a closure on web-intersection mechanism: cross-web crossings produce overdetermined identities (observable), but the underlying mechanism is multi-week Lyra Task #243 work

Tier discipline (per Cal Substrate-Cognition External Register methodology)

Level 1 (D-tier supportable internally): - BST primary integers exist with verified values (rank=2, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$, $N_{\max}=137$) - Each integer appears in many BST observable contexts (catalog data, ~4534 invariants) - Overdetermined identities cluster at specific cross-integer values ($Q=126$, $\chi=24$, $c_{FK}=225$)

Level 2 (I-tier working ontological hypothesis with falsifier): - “Integers ARE boundaries holding webs” ontological reading - “Webs interconnect at Type C convergence patterns” structural claim - “Substrate engineering operates at web intersections” operational claim

External register discipline (per Cal Substrate-Cognition External Register)

Internal use: Integer Web Principle as foundational ontology — acceptable freely.

External use: DEFAULT-DENY UNTIL CAL GRADE-PASS for ontological framing.

Acceptable external register (operational, falsifier-bearing): - “BST framework identifies six BST primary integers each appearing across multiple observable contexts” - “BST framework’s Cross-Classification Matrix maps 256 cells ($8 \text{ BST} \times 4 \text{ zones} \times 8 \text{ physical types}$)” - Specific integer-web examples (g-web with Casimir asymmetric, $N_{\max}$ -web with α) acceptable

Unacceptable external register: - “The integers ARE the boundaries” - “Substrate is made of integer-webs” - Any ontological-necessity claim without explicit hypothesis register

Connection to other principles

- SWPP: substrate as active information-cycle; Integer Web Principle specifies WHAT the substrate’s structural ontology IS (integer-webs operating in commitment cycles)
- Five-Absence Predictions: NO GUT, NO proton decay, etc. follow from D_{IV}^5 irreducibility; Integer Web Principle is the ontological refinement (D_{IV}^5 irreducibility = the 6 integer-webs braid uniquely)
- Substrate Closure Principle: substrate has no external referent; Integer Web Principle specifies the internal structure
- Graph Forces Principle: overdetermined-identity clustering as substrate diagnostic; Integer Web Principle explains WHY clustering happens (web intersections)
- Substrate Cognition Hypothesis (L2-cognition sub-class): cognition lives in long-distance correlation network between commitment circles; integer-edges provide the connection mechanism

Per Casey's standard

- Simple: six integers, each holds a web of operations
- Works: catalog data shows integer-appearances forming coherent contextual networks
- Hard to break: would require a BST primary with random/disconnected appearances; not currently observed
- Counter-example: none observed within current 4534-invariant catalog

Filing status

Casey-named principle FILED per Wednesday 2026-05-20 approve-all directive. L2 working-hypothesis tier with explicit falsifier. Internal register acceptable; external default-deny until Cal grade-pass per substrate-cognition external register methodology. This principle makes SWPP + Five-Absence + Graph Forces operational at deeper structural ontology level.

— Keeper, 2026-05-20 Wednesday EOD (Cal register discipline applied)